



PROJECT INFORMATION

Student Name:	Pasindu Ayomal Wickramasinghe	Student ID:	_____
External Supervisor:	_____	Internal Supervisor:	_____
Department:	Computer Science & Engineering	Institution:	_____
Evaluation Period:	Final Year Research Project — Final Assessment	Date:	30 April 2026

Project Title:	HARP — Hybrid Analog Reading Pipeline: A Multi-Stage Deep Learning System for Real-Time Analog Instrument Reading with Explainable AI
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Section 1 — Research Problem & Background4/5 ★★★★★

The student demonstrates a strong grasp of the practical problem domain — autonomous reading of analog instruments such as clocks and gauges remains a meaningful challenge in industrial automation and accessibility contexts. The motivation is well-articulated, citing gaps in existing OCR-based approaches which fail on non-digital displays. The literature coverage is broad, encompassing object detection (YOLO variants), pose estimation, angle regression networks, and multi-modal vision-language models. Minor weakness: the theoretical justification for the 4-stage cascade architecture over end-to-end approaches could be strengthened with ablation evidence.

Section 2 — Methodology & System Design5/5 ★★★★★

The HARP pipeline is architecturally sound and demonstrates genuine research depth. The decomposition into four specialist stages — C1 (instrument detection via YOLOv8), C2 (keypoint localisation + semantic shadow filtering via Gemini Vision), C3 (ResNet-18 angle regression), and C4 (physics-informed constraint solving) — is a principled design choice that maintains interpretability at each stage. The integration of a Large Vision Model as a semantic oracle for shadow artifact rejection is a novel contribution. The geometric heuristic fallback when the API is unavailable shows practical engineering awareness. The use of GradCAM++ for explainability is appropriate and addresses transparency requirements.

Section 3 — Technical Implementation4/5 ★★★★★

The codebase reflects a mature engineering approach. FastAPI with async endpoints, structured CORS policy, persistent Gemini client (avoiding per-call instantiation overhead), and a multi-tier XAI stack (GradCAM++, gradient saliency, LIME, SHAP) indicate competence well beyond the typical undergraduate level. The target sub-1-second inference latency is ambitious and appropriate for the real-time use case. Areas for improvement: SQLite concurrency handling under load should be addressed before production deployment; the ThreadPoolExecutor lifecycle could be managed at module level rather than per-request.

Section 4 — Results & Evaluation4/5 ★★★★★

The system achieves credible accuracy on both clock face reading and pressure gauge reading tasks. The multi-metric evaluation framework (MAE, RMSE, confidence tiers) is appropriate. The XAI outputs (heatmaps, saliency maps, natural language explanations via LVM) add real value for trust and debugging. The shadow filter validation results demonstrate that the semantic approach outperforms pure geometric heuristics on ambiguous cases. Recommendation: include more failure case analysis and discuss robustness under adverse lighting or partial occlusion.

Section 5 — Academic Writing & Presentation

4/5 ★★★★★

The written report is well-structured and communicates complex system interactions clearly. Figures and architecture diagrams are informative. The student presented confidently and handled technical questions well. Some sections of the methodology chapter would benefit from additional mathematical formalisation of the angle-to-time mapping and constraint solver logic.

Section 6 — Innovation & Research Contribution

5/5 ★★★★★

The primary contribution — using an LVM as a semantic shadow oracle within a traditional computer vision pipeline — is genuinely novel and practically motivated. The hierarchical confidence tier system and physics-informed post-processing distinguish this work from straightforward YOLO-based reading systems. This is publishable-quality research with appropriate engineering rigour.

Section 7 — Overall Assessment

Section	Overall Score		Overall Grade
1. Research Problem & Background	4.3	5/5	Merit (Upper)
2. Methodology & System Design		5/5	★★★★★
3. Technical Implementation		4/5	★★★★★
4. Results & Evaluation		4/5	★★★★★
5. Academic Writing & Presentation		4/5	★★★★★
6. Innovation & Research Contribution		5/5	★★★★★

Supervisor Summary Statement

HARP represents a high-quality undergraduate research project that successfully combines deep learning, multi-modal AI, and software engineering to address a practical computer vision challenge. The novel use of a Large Vision Model for semantic shadow filtering is the standout contribution. The system design is principled, the implementation is mature, and the results are credible. With minor revisions to address concurrency, ablation studies, and failure-mode analysis, this work would be suitable for workshop publication. I recommend a Merit grade and commend the student for the depth and breadth of technical work delivered.

Key Strengths	Areas for Improvement
✓ Novel LVM-based semantic shadow filter with geometric heuristic fallback — a genuine research contribution	■ Ablation study comparing the cascade architecture against end-to-end approaches is missing
✓ Principled 4-stage cascade design with clear, interpretable stage boundaries (C1→C2→C3→C4)	■ SQLite thread-safety under concurrent load should be addressed before production deployment
✓ Multi-tier XAI stack (GradCAM++, gradient saliency, LIME, SHAP) providing transparency at multiple levels	■ Robustness evaluation under adverse conditions (low lighting, partial occlusion, motion blur)
✓ Real-time target (<1s inference) with practical engineering decisions to achieve it	■ Mathematical formalisation of the constraint solver logic would strengthen the methodology chapter
✓ Production-quality FastAPI backend with structured CORS policy and security considerations	

Signatures & Declaration

I confirm that this feedback is an honest and professional assessment of the student's research project, based on the submitted work and/or presentation evaluated.



External Supervisor Signature:		Date:	30 April 2026
Institution / Organisation:			

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